

ClassScribe — What We Are Doing, and Why It Works

Presentation 1 · Research half of the final-year project Syllabus-Grounded Contextual Decoding (SGCD) for Hindi–English code-switched lecture transcription.

Repository: [research/](#) (the science) · [backend/](#) (the working system built on it)

1. The one-line version

Indian lecturers teach in mixed Hindi–English. Off-the-shelf speech recognisers transcribe such lectures badly — especially the technical words a student actually needs. **We make an existing speech model transcribe these lectures better without retraining it, by feeding it the course syllabus in the exact form the model's context channel was built to accept.** We then measured, with proper controls, how much this helps and *why* it helps.

The research is not "we wrote a good prompt." It is: **we identified where a pre-trained decoder loses information, designed an inference-time correction for it, and ran a pre-registered controlled experiment that isolates the mechanism** — including a control that proved our own original hypothesis wrong, which we report honestly.

2. The problem, in three parts

A sentence from a real Indian classroom:

"इस tutorial में हम file permissions समझेंगे"

Notice: the **grammar is Hindi**, the **content words are English**. This is called **code-switching** — Hindi is the *matrix language*, English is the *embedded language*.

Three things go wrong when you feed this to a general speech model like Whisper:

#	Failure	What it looks like	Why it matters
1	Technical-term error	file permissions → misheard entirely	Those are exactly the words a student needs for exams
2	Orthographic / script error	model writes स्क्रीन where the correct form is screen	The transcript becomes unsearchable — you cannot Ctrl+F for a term written in the wrong script
3	Degenerate repetition	model loops one phrase until the token limit	One bad utterance destroys a whole passage

Failure 2 is the interesting one. It is **systematic, not random** — meaning it is in principle *correctable by conditioning*, without touching the model weights. That is our opening.

3. The idea

Every college already has a document that lists, **before the lecture happens**, exactly which topics will be taught and in which vocabulary: **the course syllabus**. It is free, it is written by the instructor, and it needs zero annotation.

Nobody uses it for ASR.

So: can we hand the syllabus to the decoder as auxiliary context at inference time?

Whisper (and every encoder–decoder speech model) has a slot for this — the `initial_prompt` / decoder context channel. It was trained to receive **the transcript of the previous chunk of speech**. That single fact drives every design decision we made.

4. Why the obvious approach fails (and why this is our contribution)

The obvious thing is to dump the syllabus keywords in: `chmod, permission, execute, owner, ...`

This is the published state of the art for training-free biasing, and it is known to backfire (Sun et al., Interspeech 2023). **We reproduced that failure in a new language setting and then fixed it.**

What happens with a keyword list:

- Error on the **listed** terms drops by about half. ✓
- Error on **every other word** goes up. ✗
- Ordinary words outnumber technical words in any sentence, so **overall accuracy gets worse**.

The diagnosis — distribution mismatch. A comma-separated term list is not something the context channel has ever seen during training. It is *out of distribution*. The decoder is destabilised: it starts inserting, looping, and drifting.

Our fix — change the representation, not the content. We take the *same* syllabus content and render it as **fluent code-mixed narration** — a sentence a lecturer would actually speak:

"इस tutorial में हम पहला c program लिखना सीखेंगे। हम main function बनाएंगे, printf से output दिखाएंगे, फिर gcc से compile करके program run करेंगे।"

Now the context channel receives something shaped exactly like what it was trained on. Same information. Completely different outcome.

This is the headline scientific claim: representation format, not information content, determines whether auxiliary context helps or harms a pre-trained decoder.

5. The method — SGCD, four moving parts

5.1 Syllabus → conditioning units

Each course becomes a list of **units**, each a triple `(title, prose, keywords)`. The `prose` field is engineered under three hard constraints derived from how the context channel behaves:

- **Register & orthography** — flowing instructional narration, Hindi function words in **Devanagari**, English technical terms in **Latin**. This teaches the model the corpus's **dual-script output convention**, not just the vocabulary.
- **Terminal weighting** — later context tokens influence generation more, so the most distinctive terminology is placed **at the end**, and multiple units are ordered **ascending by relevance** so the best one lands last.
- **Token budget** — measured with Whisper's own BPE tokeniser, code-mixed text costs **~3 tokens per word**. Each unit is 60–105 tokens; the assembled context is **left-truncated at 200 tokens** so the high-influence tail

survives.

The `keywords` are kept, but used **only as a retrieval signal** — they are never sent to the decoder. That is deliberate: keywords are what caused the failure in §4.

5.2 Two-pass contextual retrieval

Which syllabus units should we supply? That needs to know the topic, which needs the transcript — circular. We break the circle:

```
audio → PASS 1: decode with no context      → hypothesis h (noisy, but lexically indicative)
      → retrieve k=3 syllabus units using h as the query
      → PASS 2: decode again, conditioned on those units as prose
```

Retrieval is **character n-gram TF-IDF (n = 3–5), cosine similarity**. Character n-grams rather than word tokens, for two setting-specific reasons: code-mixed text has inconsistent transliteration so word matching fails on spelling variants, and the pass-1 hypothesis is noisy by construction so we need a representation that **degrades gracefully**. It is sparse — no model download, ~1 ms per query.

Cost: exactly one supplementary decode. Pass 1 is the hypothesis the baseline system produces anyway.

5.3 Stability safeguard

Conditioning is not uniformly good per utterance. Error analysis showed it does two opposite things: it **repairs** repetition loops in unconditioned output, and it also **causes** them — the model continues the syllabus narration's register instead of stopping.

So we accept the conditioned output only if decoding stayed stable. We revert to the pass-1 output if **any** of:

Signal	Rule	Catches
Confidence drop	<code>logprob_cond < logprob_base - δ</code> ($\delta = 0.25$)	model became unsure
Compression ratio	<code>ratio > ρ</code> ($\rho = 2.0$)	repetitive text compresses well → loop detector
Length runaway	<code>words_cond > λ × words_base</code> ($\lambda = 1.5$)	generation not terminating

Thresholds were fitted on a development split and **frozen before the test run**.

5.4 Span length — a precondition, not a tuning knob

This turned out to be load-bearing. At the corpus's native **5.7 s** utterances, conditioning *hurts* (+5.11 WER) and needs the safeguard just to break even. At **26.2 s** spans it helps outright (−6.23 WER, safeguard not needed). Longer spans give the context more material to act on. Whisper's encoder window is 30 s, so the production system segments at **25 s** — inside the window, in the regime where the method works.

6. How we tested it — this is what makes it research

Anyone can report a number that looks good. The design below is what makes the number trustworthy, and it is worth showing your examiners.

- **Dataset:** MUCS 2021 subtask-2 Hindi–English (**OpenSLR SLR104**, CC BY-SA 4.0) — spoken technical tutorials, so a syllabus is a *natural* context source, not a contrived one. 30 lectures, 3073 utterances after filtering, **5.06**

hours. Test tarball only — the training split is never downloaded, so the corpus's known 33.9% train/test overlap does not apply to us.

- **Zero-shot.** Nothing is trained or fine-tuned. Every gain is inference-time.
- **Lecture-disjoint split.** 30% of *lectures* → development (tuning), 70% → test. Splitting by lecture, not by utterance, prevents same-speaker leakage.
- **Pre-registration.** Five hypotheses with numeric pass criteria were **committed to git before the test split was ever decoded** (`sgcd/PREREGISTRATION.md`), including an explicit promise to report the result either way.
- **The test set is decoded exactly once**, after the configuration was frozen.
- **Leakage guards in code.** Syllabi were written from lecture *titles* only; the utterances those titles came from are programmatically excluded from scoring. An assertion raises if a reference transcript is ever used as a retrieval query outside the labelled oracle condition.
- **Append-only run log** (`sgcd/RUNLOG.md`) — every run, every number, including the ones that went against us.
- **One-command replication** — `make_comparison.py --verify` re-checks every published number *and* the nine qualitative claims, and exits non-zero on mismatch.

The conditions (this is the experimental design)

ID	Context given to the decoder	Role
A / C0	nothing	baseline
G / C1	one generic code-mixed sentence, zero course content	isolates <i>pure format</i> effect
B / C2	syllabus keywords, comma-separated	the naive method — the known failure
C / C3	whole syllabus as prose	tests the rendering hypothesis
D / C4	k=3 retrieved units as prose = SGCD	proposed method
E / C5	retrieved from a different course	the load-bearing control
F / C6	retrieved using the ground truth	oracle topline, always labelled
C7	D + stability safeguard	full system

Condition E is the intellectually important one. Same length, same register, same construction — only the *content* is wrong. If a wrong-course syllabus helps just as much as the right one, then the gain is **not** coming from syllabus knowledge. Most published contextual-biasing work has no such control.

Metrics (know these for the viva)

Metric	What it measures
WER	word error rate, lower better. Can exceed 100% when the model inserts extra words
K-WER	error on words that appear in the syllabus terminology — <i>did we fix the technical terms?</i>
U-WER	error on all other words — <i>did we break everything else?</i>
Script fidelity	% of Latin-script reference words recovered in Latin — measures the dual-script convention
WER-sa	transliteration-tolerant WER: both scripts reduced to a consonant skeleton (<code>screen</code> / <code>स्क्रीन</code> → <code>skrn</code>). A <i>lower bound</i> on error, reported beside strict WER, never instead of it
95% CI	paired bootstrap, 10,000 resamples, seed 1337 — tells us the difference is real, not noise

Splitting WER into **K-WER vs U-WER** is the diagnostic that makes the whole paper work. Aggregate WER alone cannot tell you "helped the terms, wrecked the rest" — that pattern is invisible until you split it.

7. Results

7.1 Lecture-length spans (100 spans, mean 26.2 s) — the realistic setting

Method	WER ↓	Δ vs A [95% CI]	Technical-term error ↓	Everything-else error ↓	Script fidelity ↑
A — no context	43.46	—	48.04	34.64	48.0%
B — terminology list	62.86	+19.39 [+7.13, +33.06]	23.00	46.42	70.3%
C — syllabus as narration	37.23	−6.23 [−10.59, −1.97]	21.38	32.42	71.6%

Read it like this: the keyword list (B) halves the error on the words it targets — and still makes the transcript worse overall, by 19 points, because it damages everything else. The *identical content* written as narration (C) gets the **same** terminology gain (−55.5% relative), and non-term error goes **down** instead of up. Every confidence interval here excludes zero. Direct B-vs-C contrast: **−25.62** [−39.28, −13.50] in favour of narration.

7.2 Short utterance-level segments (150 utterances, mean 5.7 s)

Method	WER ↓	Δ vs A	K-WER	U-WER	Script fid.
A — no context	85.69	—	65.53	43.50	32.6%
G — generic sentence, no content	102.32	+16.63	55.83	44.38	41.6%
B — terminology list	136.22	+50.53 [+16.55, +90.94]	31.55	53.18	60.9%
C — narration	99.35	+13.66	33.50	41.68	60.0%
D — SGCD (retrieved)	90.80	+5.11	35.44	41.95	58.8%
E — mismatched syllabus	87.71	+2.02	35.92	40.19	58.6%
C7 — SGCD + safeguard	71.50	−14.19 [−26.55, −4.79]	35.44	41.14	58.1%

Why the absolute numbers are high — have this answer ready. Three documented causes, all applying equally to every condition, so the *comparison* is unaffected: (i) 99.6% of the corpus's segment boundaries are rounded to whole seconds, so fragments of neighbouring utterances bleed into every clip; (ii) we deliberately disabled Whisper's temperature-fallback loop so repetition instability stays *measurable* instead of being silently patched; (iii) strict WER punishes script mismatch, which is why WER-sa is reported alongside (71.87 at baseline). **These numbers are not comparable to MUCS leaderboard figures, and we say so in the paper.** Note that baseline WER halves — 85.69 → 43.46 — purely by moving to realistic span lengths, which confirms the short-utterance numbers were inflated by segmentation artefacts and not by our method.

7.3 Pre-registered hypotheses — the outcome

ID	Claim	Result	Verdict
H1	Full system beats baseline on WER	−14.19, CI excludes 0	PASS
H2	Syllabus context improves technical-term recognition	K-WER 65.53 → 35.44 = 46% relative	PASS
H3	The gain is content-specific	matched vs mismatched: +3.09, CI spans 0	FAIL
H4	Prose beats keyword list	−36.88, CI excludes 0	PASS
H5	Retrieval beats whole-syllabus prompting	−8.55, CI spans 0	directional, not significant

8. The honest finding — say this out loud, it is a strength

H3 failed, and we report it as the central caveat of the paper.

A syllabus from the **wrong course** recovers roughly **81%** of the improvement. The oracle-retrieval topline is no better than ordinary retrieval. On lecture-length spans, of SGCD's 4.83-point gain, the mismatched syllabus already delivers 3.91 — content contributes **0.92 points (~19%)**, with a **confidence interval that still includes zero**.

So what is actually happening? Look at the script-fidelity column across conditions: 32.6% (no context) → 41.6% (one generic sentence, *zero* course content) → ~60% (any syllabus, right or wrong). The context is teaching the model the corpus's **output convention** — *write English technical terms in Latin, Hindi in Devanagari* — far more than it is telling the model which specific words to expect. This is a **format / register effect**, not semantic grounding.

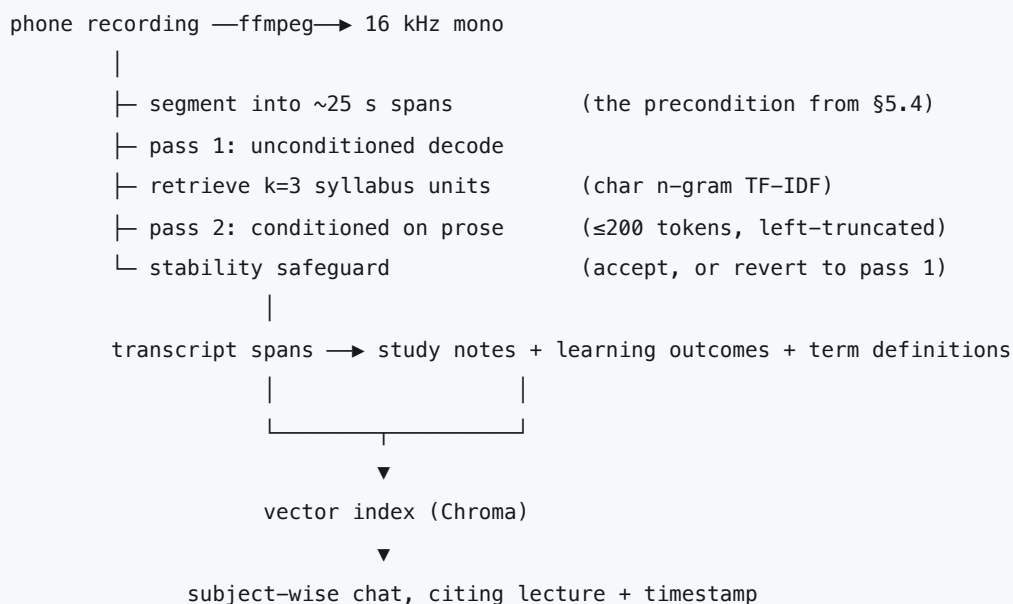
Three reasons this makes the work stronger, not weaker:

1. **Our controls caught it.** Without conditions E and G we would have published a content result that was false. That is exactly what controls are for, and most contextual-biasing papers do not include them.
2. **It is practically good news.** Deployment does **not** need an accurate per-lecture syllabus to get most of the benefit — a huge lowering of the adoption barrier for a college.
3. **It defines the follow-up precisely.** Resolving a 0.92-point effect at the observed variance needs ~700 spans — the full 5-hour partition. That is roughly 20 minutes of compute per condition. Named, costed, and achievable.

Cross-model check. Repeating everything on a smaller Whisper checkpoint preserves the ordering (enumeration catastrophic, narration beneficial) and adds a nuance: the small model is nearly *script-blind* (1.6% script fidelity, 99.5% of technical terms wrong), and there the **matched** syllabus does beat the mismatched by 17.52 points. Plausible reading, offered as hypothesis not finding: **content matters when the model is too weak to produce the terminology at all; once the model is strong enough to get the words, only the convention is still missing**. We also found the safeguard thresholds do **not** transfer across checkpoints — on the small model they fired on 46% of utterances and made things worse. Stated as a real limitation of the method.

9. From research to product — the `backend/` folder

The research proves the method. The backend is that method running as a real system for students.



Built as a **FastAPI** service with SQLAlchemy persistence, a background job queue for long operations, and a Chroma vector store. Key points to mention:

- **Syllabus ingestion is a rewrite, not text extraction.** A syllabus PDF is a terminology list in prose clothing — bullets, module tables, comma-separated topics. Fed to the decoder in that shape it reproduces the failure mode from §4. So the PDF is passed through an LLM that rewrites each unit into 20–35 words of code-mixed narration under the four rules from §5.1. **This is the research finding made operational.**
- **Notes and learning outcomes** are generated per topic, with terminology definitions, outcomes tagged by Bloom level.
- **Syllabus coverage tracking comes free.** Retrieval already selected a syllabus unit per span during decoding, so "which units have been taught, which are outstanding" is a database group-by, not a separate classification pass.
- **Query routing.** A question is classified first (`lookup` , `explain` , `summary` , `compare` , `quiz` , `outline` , `coverage` , `smalltalk`), because a definition lookup wants two sentences from one span while a revision request wants the notes for a whole unit. Coverage questions are answered from the database with no retrieval at all — it is a computable fact.
- **Every answer cites** `[n]` → `{lecture_id, timestamp}` , so any claim can be checked against the recording.
- **Runs locally.** Transcription is on-device at 5.5–15.7× real time on a consumer laptop, so classroom audio — which includes students' voices — need not leave institutional infrastructure. That answers the data-governance objection to Otter.ai and cloud captioning.
- **27 automated tests**, ~16 s, no model download and no API key needed — covering segmentation bounds, retrieval ranking under a noisy first pass, prompt construction and left-truncation, all three safeguard triggers, and a full upload → decode → notes → coverage run with the model stubbed.

10. What is new here (the defensible / patentable core)

1. **A representation transform:** syllabus content converted into register- and orthography-matched code-mixed narration, engineered against a decoder context channel's known behaviour (terminal token weighting, bounded budget, left-truncation preserving the tail).
2. **A two-pass conditioning loop that costs one extra decode:** the unconditioned first pass doubles as the retrieval query, so context selection is free.

3. **Sparse character n-gram retrieval chosen for code-mixed noise tolerance** — no neural encoder, no model download, millisecond latency.
4. **A three-signal stability safeguard** that keeps conditioning's repairs and discards its regressions.
5. **Span-length as an explicit operating precondition**, with measured evidence for where the method works and where it inverts.
6. **Retrieval-derived syllabus attribution**, which yields automatic syllabus-coverage tracking as a by-product of decoding rather than as a separate step.
7. **A controlled evaluation protocol** — mismatched-syllabus and contentless-register controls — that separates *format* conditioning from *semantic* conditioning. To our knowledge this separation has not been isolated for code-switched lecture ASR.

Point 7 is worth emphasising: the mismatched control is a **methodological contribution** in its own right. Evaluations omitting it cannot tell semantic grounding apart from register conditioning.

11. Limitations we state ourselves

- One corpus, and it is **scripted** tutorial narration — more fluent than spontaneous classroom speech.
- Samples of 150 utterances / 100 spans are **small**; the content-specificity question in particular is **underpowered**.
- Syllabi were **generated from lecture titles** by a language model, not taken from published institutional outlines. No scored utterance contributed to them, but it remains a confound.
- Safeguard thresholds are **model-specific** and must be refitted per checkpoint.
- Absolute WER is inflated by corpus segmentation artefacts and is **not comparable** to published MUCS figures.

Next steps: re-run content specificity at $n \approx 700$ spans (the full partition, ~ 20 min/condition); evaluate on spontaneous classroom speech; replace the safeguard with a threshold-free criterion that needs no per-checkpoint calibration; and evaluate the notes and chat against actual learning outcomes with students — because transcription accuracy is a *proxy for*, not a *measure of*, educational usefulness.

12. Three sentences to close the presentation

1. We took a pre-trained speech model and made it substantially better at Hindi–English lecture transcription **without training anything** — 6.23 points of WER, 55.5% relative reduction in technical-term error, and script fidelity from 48% to 72% — by supplying the course syllabus in the form the decoder's context channel was designed to receive.
 2. We showed that the *representation* of auxiliary context, not its information content, is what decides whether it helps or harms — a terminology list of the very same words makes the transcript **19 points worse**.
 3. And because we built the right controls, we can say precisely *what* is doing the work: the model is mainly being taught the correct output convention, not the course content — a finding that lowers the deployment barrier and tells us exactly which experiment to run next.
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Quick glossary for the viva

ASR automatic speech recognition · **Code-switching** alternating two languages inside one utterance · **Matrix / embedded language** the grammatical frame vs the inserted vocabulary · **WER** word error rate · **K-WER / U-WER**

error on syllabus-keyword vs all other reference words · **Script fidelity** fraction of Latin-script reference words recovered in Latin · **Contextual biasing** steering recognition toward known relevant terms · **Decoder context channel** / **initial_prompt** the slot Whisper accepts prior text in · **Distribution mismatch** giving a model input unlike anything in its training data · **Zero-shot** no training or fine-tuning on the target data · **TF-IDF** / **character n-gram** sparse text similarity robust to spelling variation · **Terminal weighting** later context tokens influence generation more · **Compression ratio** a repetition detector — looped text compresses unusually well · **Paired bootstrap CI** resampling test for whether a difference is real · **Pre-registration** committing hypotheses and pass criteria before seeing the test data · **Lecture-disjoint split** splitting by recording so no speaker appears on both sides